Aadi Sinha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.77
Delhi Private School Dubai, UAE	Class XII (CBSE)	2023	97.60%
Delhi Private School Dubai, UAE	Class X (CBSE)	2021	97.80%

SCHOLASTIC ACHIEVEMENTS

International Mathematical Olympiad (IMO)

United Arab Emirates | *2022 & 2023*

- Represented the UAE at the International Mathematical Olympiad in 2022 and 2023, earning an Honourable Mention in both years and competing among top mathematics students globally.
- Scored 13/42 at the 2023 IMO, contributing approximately 36% of the UAE team's 36-point total through rigorous problem-solving across advanced mathematical disciplines.

SKILLS

Programming Skills: C/C++, Java, Python, SQL.

RELEVANT COURSEWORK

Mathematics Courses: Probability, Statistics, Number Theory, Calculus, Differential Equations, Complex Variables, Transform Techniques, Numerical Methods.

Machine Learning: Machine Learning for Mechanical Engineers; Stanford CS229 - Machine Learning (Andrew Ng).

FREELANCE EXPERIENCE

Research Analyst | *Mathematics, Latex*

Turing AI | *Remote*

- Designed and proposed challenging IMO-style mathematics problems to probe the limits of a state-of-the-art AI reasoning model, identifying problems the model could not reliably solve and expanding the difficulty of its evaluation dataset.
- Developed rigorous human-verified solutions and reasoning paths for selected problems, ensuring mathematical correctness and clarity while providing high-quality training and evaluation data to improve the model's reasoning and problem-solving capabilities.
- Collaborated with a US-based team supporting a European client, delivering solutions and incorporating feedback to meet client-specific quality standards while contributing to the development and evaluation of an AI model specialized in advanced mathematical reasoning.

Aerospace Engineering Expert | *Slack, Insightful*

Mercor | *Remote*

- Developed challenging aerospace evaluation problems using technical artifacts--including graphs, plots, diagrams, engineering drawings, and flow representations--to identify failure modes in current AI models.
- Designed problems that exposed limitations in AI reasoning, ensuring questions required genuine interpretation of aerospace artifacts rather than pattern matching or straightforward information retrieval.
- Produced independent, technically rigorous solutions and supporting reasoning for each problem, validating the expected solution against the provided aerospace artifact.
- Collaborated remotely with a globally distributed team across multiple time zones, communicating with project managers, resolving technical ambiguities, and participating in weekly project meetings.

WORK EXPERIENCE

Supply Chain Management Intern | *MS Office, Python, MySQL, Power Automate*

Larsen & Toubro | *Chennai, Tamil Nadu*

- Analyzed procurement data across multiple Independent Companies (ICs) and projects to identify common item groups suitable for centralized procurement and rate-contract (RC) consolidation; flagged high-potential items and reported recommendations to respective procurement teams.
- Developed an engineering-backed should-cost model for heavy equipment procured by the CAPEX team, incorporating technical specifications and cost drivers to establish data-driven benchmarks for supplier negotiations.
- Supported procurement negotiations using should-cost benchmarks, contributing to 12-20% price reductions against the previous fiscal year's procurement benchmarks for selected equipment.
- Conducted market research on the impact of geopolitical disruptions in West Asia on ductile iron (DI) pipe prices, mapping raw-material and supply-chain dependencies to assess potential procurement risks and price movements.
- Mapped item groups requiring manual procurement outside L&T's digital procurement portal, identifying commonly purchased items suitable for new item-code creation and process digitization to improve procurement coverage.

PROJECTS

Propeller Design | *MATLAB, SolidWorks, Ansys Fluent*

- Designed a compact propeller system for a Short Takeoff and Landing (STOL) aircraft, targeting high thrust-to-weight ratio and lift augmentation while minimizing propeller size.
- Developed propeller geometry using BEMT, optimizing spanwise twist, chord distribution, and airfoil selection for operation at 30,000 RPM, with QPROP predicting 842 W propeller power.
- Validated the propeller design using ANSYS Virtual Blade Model (VBM), achieving 12 N thrust and a 0.65 thrust-to-weight ratio at a 20 m/s freestream velocity, with a relatively uniform 51-56 m/s wake velocity.
- Presented the aerodynamic design methodology, simulation results, and performance analysis to a panel of judges at IIT Patna, contributing to a 4th place finish among IIT teams in the Inter-IIT competition.

Topology Optimization | *Altair Inspire, Fusion 360, Meshmixer*

- Optimized structural components of a FIRST Robotics Competition (FRC) robot by identifying over-designed/high-mass regions and defining realistic load cases at joints and externally loaded interfaces.
- Performed topology optimization on multiple components, including rotating assemblies, accounting for load variations and stresses throughout the operating rotation to reduce unnecessary material while ensuring safe life.
- Redesigned optimized geometries in CAD to eliminate stress concentrations induced by sharp features, holes, and unfavorable geometry, converting topology-optimized results into manufacturable components.
- Reduced component mass by 26%-52%, contributing to lightweight structural design for a high-speed robotic system.
- Secured 2nd place globally in the Altair Global Student Contest for the structural optimization study.

Reduced-Order Modelling of Flapping-Wing Aerodynamics | *PIV, MATLAB, 3D Printing*

- Investigating unsteady flow structures and vortex formation around a 100 mm-span flapping wing to study aerodynamic effects not captured by a quasi-steady model.
- Designing a 2-DOF wing-motion mechanism to reproduce flapping and wing-rotation kinematics, actuated by two stepper motors and fabricated using 3D printing for PIV experiments.
- Targeting 40 Hz flapping frequency with approximately ±45° flapping amplitude, while evaluating actuator response and mechanical feasibility for reproducing the prescribed wing kinematics.
- Developing a PIV-based experimental methodology to characterize the evolution of streamlines and vortical structures over the flapping cycle and use the measurements to inform additional terms in a reduced-order aerodynamic model for an ornithopter.